

Sri Hari Sirisipalli

AI & Mechanical Systems Engineer | GenAI, Computer Vision, CFD/FEA, Backend & Cloud Systems

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Profile

AI and Mechanical Systems Engineer with experience building GenAI applications, computer vision pipelines, engineering-focused AI systems, digital twin workflows, simulation automation systems, and cloud-native backend infrastructure. Skilled in applying LLMs, RAG, embedding pipelines, computer vision, time-series modeling, anomaly detection, surrogate modeling, and retrieval systems to mechanical, offshore, and domain-specific engineering problems. Background includes CFD methods, FEA workflows, CAD/CAE automation, ANSYS APDL, MOSES, FreeCAD, structural and modal analysis, aerodynamic flow simulation, marine/offshore response modeling, and large-scale visual similarity search. Experienced in Python, FastAPI, Django, AWS, Terraform, Docker, WebSockets, vector databases, data pipelines, API integration, and production deployment workflows. Strong foundation in mechanical engineering, fluid mechanics, structural analysis, numerical methods, GenAI systems, computer vision, backend engineering, cloud infrastructure, and DevOps for simulation-backed and AI-powered engineering applications.

Professional Experience

AI & Mechanical Systems Engineer,
Independent Consulting & R&D

06/2023 – Present

Built AI, GenAI, computer vision, simulation automation, digital twin, and engineering software systems across offshore, naval, industrial, agritech, and document intelligence domains.

- **Offshore Riser Digital Twin:** Developed ML-based surrogate modeling workflows for offshore riser fatigue, vessel motion response, wave height estimation, and wave heading prediction.
- **CAD/FEA Automation:** Built Python, ANSYS APDL, FreeCAD, and MOSES workflows for parametric geometry generation, structural/modal analysis, batch execution, and results extraction.
- **CFD Flow Simulation:** Applied CFD methods including geometry preparation, meshing, boundary condition setup, drag polar analysis, aerodynamic coefficient extraction, and flow visualization.
- **PyNAOS Offshore FEA Engine:** Developed a Python-based offshore structural analysis engine and GUI for static, dynamic, eigenvalue, nonlinear, buckling, wave loading, and soil-pile workflows.
- **Naval Architecture Computation Library:** Built Python modules for hull geometry, hydrostatics, stability, resistance, propulsion, seakeeping, maneuvering, strength, and rule-based checks.
- **Naval Machine Fault Detection:** Built vibration-based anomaly detection and classification pipelines using time-domain features, frequency-domain features, PCA, clustering, and ML models.
- **Signal Processing & Anomaly Detection:** Developed unsupervised fault detection workflows using normal-operation baselines, distance thresholds, and clustering-based anomaly scoring.
- **Computer Vision Retrieval System:** Architected visual biometric retrieval pipelines using YOLOv8, OpenCV, SimCLR, FAISS, multi-view embeddings, and ranking fusion.
- **GenAI / Offline AI Avatar:** Built an offline conversational AI system using Whisper, Ollama, LangChain, ChromaDB, Silero TTS, FastAPI, WebSockets, React, and Three.js.
- **RAG over Technical Reports:** Implemented retrieval-augmented generation pipelines over internal technical documents using chunking, metadata filtering, FAISS indexing, and FastAPI services.
- **Ship Recycling Expert System:** Built a Django-based expert system for ship recycling plans, hazardous material identification, dismantling workflows, compliance checks, risk assessment, and PDF reports.
- **Backend & Cloud Systems:** Developed Python backend services, data pipelines, API integrations, vector search systems, deployment workflows, and infrastructure automation for AI applications.

Software Engineer – LLM & ML Infrastructure, Pangeon

03/2024 – Present | Contract

Lead development of scalable embedding-driven retrieval and LLM-based evaluation systems for large-scale patent similarity analysis.

- Developed and deployed high-throughput embedding pipelines integrated with LLM semantic scoring for automated prior-art ranking.
- Support 1,000+ patent similarity evaluations daily, enabling scalable semantic comparison across large patent datasets.
- Re-architected compute-intensive workloads from EC2 to AWS Lambda, reducing infrastructure costs by 30% and improving batch completion speed by 5–7x.
- Built asynchronous processing and multiprocessing frameworks to increase similarity evaluation throughput.
- Provisioned reproducible AWS infrastructure using Terraform to enable consistent cross-account deployments.

Data Engineer, Sas2Py

03/2023 – 06/2023 | Full-Time

- Led migration of legacy SAS data pipelines to PySpark, redesigning batch workflows for distributed execution and improved scalability.
- Built automated validation framework ensuring >99% functional parity between SAS and PySpark outputs across production workloads.
- Modeled job dependency graphs to eliminate scheduling bottlenecks and optimize distributed execution efficiency.

Machine Learning Intern, Corteva Agriscience

07/2022 – 12/2022 | Internship

- Converted TensorFlow and PyTorch models to ONNX for cross-platform deployment compatibility.
- Optimized inference performance using structured pruning and post-training quantization.
- Validated numerical consistency across model formats to ensure reliable production deployment.

Software Engineer Intern, Dojima Networks

06/2022 – 12/2022 | Internship

- Integrated Polkadot ecosystem components to enable cross-chain interoperability within Dojima’s blockchain infrastructure.
- Implemented cross-chain communication workflows for transaction routing and state synchronization.
- Deployed Prometheus–Grafana monitoring stack to provide real-time API observability and system health tracking.

Core Skills

AI, GenAI & ML: LLMs, RAG, Embedding Pipelines, Vector Search, Digital Twins, Surrogate Modeling, Deep Learning, Feature Engineering, Time-Series ML, Anomaly Detection

Computer Vision & Retrieval: YOLOv8, OpenCV, SimCLR, ResNet, FAISS, Visual Similarity Search, Multi-View Embeddings, Ranking Fusion, Image Preprocessing

Mechanical Simulation & Engineering: CFD, FEA, Structural Analysis, Modal Analysis, Fluid Mechanics, Hydrodynamics, Offshore Systems, Marine Simulation, Fatigue Prediction, Numerical Methods

CAD/CAE & Simulation Automation: ANSYS APDL, MOSES, FreeCAD, CAD Automation, Parametric Modeling, Mesh Generation, Batch Simulation, Results Extraction, Engineering Post-Processing

Backend, Cloud & DevOps: Python, FastAPI, Django, AWS EC2, AWS Lambda, SQS, IAM, Terraform, Docker, CI/CD, WebSockets, API Integration

Data Engineering: PySpark, pandas, NumPy, SQL, ETL Pipelines, Time-Series Processing, Experiment Tracking, Data Validation

Programming: Python, SQL, JavaScript, Bash

Education

Bachelor of Technology (B.Tech) – Mechanical Engineering,
Mahindra University, Hyderabad

08/2018 – 06/2022 | Hyderabad

- CGPA: 7.5